

Client Implementation Review

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Project ID: CMPG325-2026-124

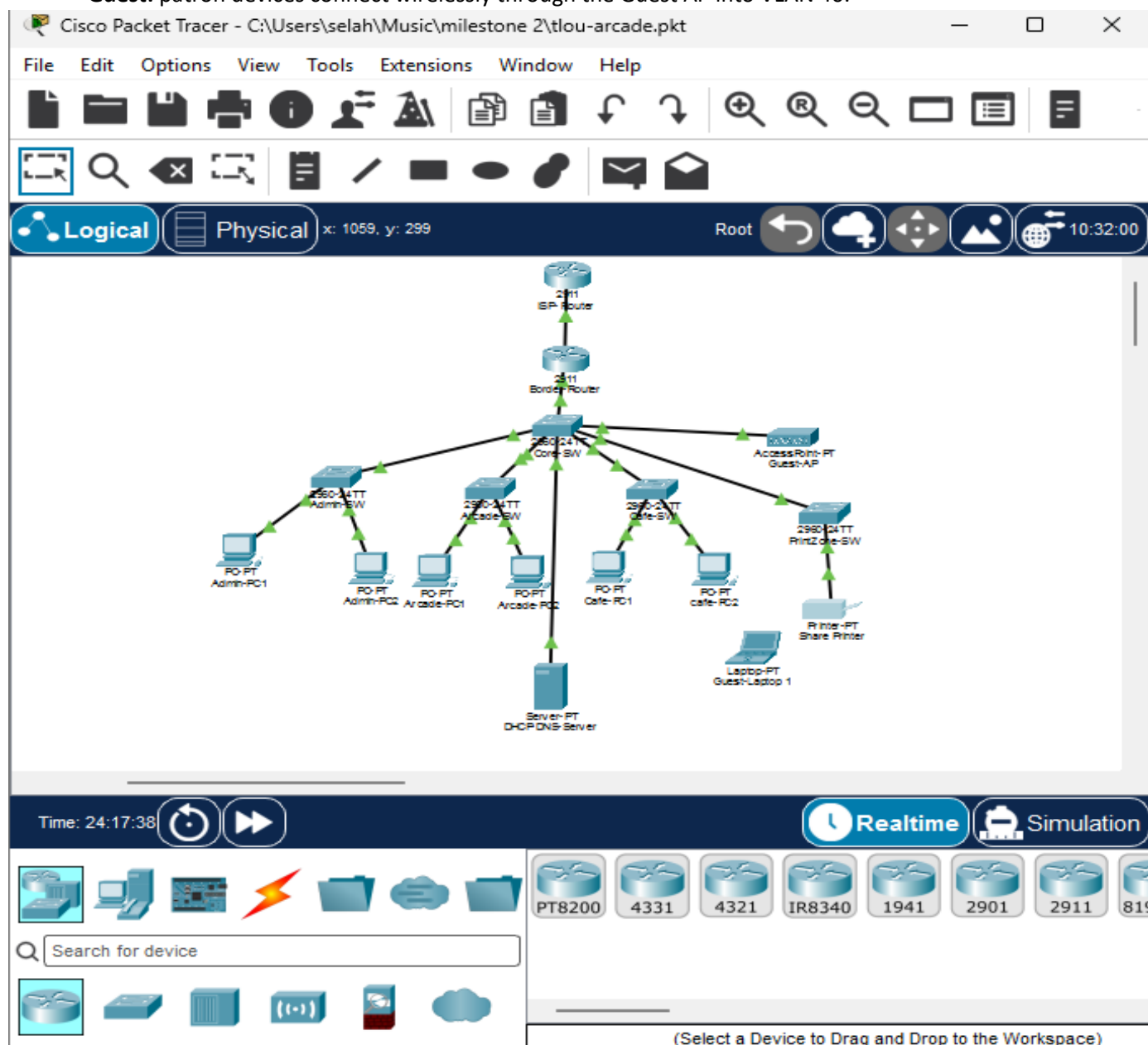
Client ID: CLI-124

Client: Tlou Amusement Arcade & Family Fun Centre (Vryburg)

1. Working Packet Tracer File

The network follows the Milestone 1 design: 192.168.54.0/24 is split by VLSM into six VLANs (10 Admin, 20 Arcade, 30 Cafe, 40 Guest, 50 Print-zone, 60 Services) plus a /30 WAN link. BORDER-R1 (Cisco 2911) provides inter-VLAN routing, NAT and DHCP relay.

- **LAN trunk:** BORDER-R1 Gi0/0 carries sub-interfaces .10 to .60 to CORE-SW, which trunks to the five access switches.
- **WAN:** BORDER-R1 Gi0/1 (192.168.54.145/30) links to ISP-R (192.168.54.146/30), which also holds a loopback 8.8.8.8 as a simulated internet host.
- **Services:** the DHCP/DNS server sits at 192.168.54.130 in VLAN 60; the printer in VLAN 50 uses static IPv4 and IPv6 addresses.
- **Guest:** patron devices connect wirelessly through the Guest AP into VLAN 40.



2. Assigned Feature Implemented: IPv6 Dual-Stack Addressing and Routing

Every VLAN runs IPv4 and IPv6 side by side on the same router sub-interface, and both are routed natively with no tunnelling.

What was configured

- `ipv6 unicast-routing` enabled globally on BORDER-R1.
- One /64 per VLAN from the ULA prefix `FD54:ACDE:2026::/48`; the router takes `::1` on each sub-interface next to its IPv4 gateway.
- Hosts build their address with SLAAC from router advertisements. Admin, Arcade and Cafe also use stateless DHCPv6 for DNS (`ipv6 nd other-config-flag` plus pool LAN-V6). Guest uses SLAAC only.
- The printer and server use static IPv6; the WAN link uses `FD54:ACDE:2026:FFFF::/64` with a static IPv6 default route to ISP-R.
- IPv6 ACLs GUEST6-IN and TO-PRINTZONE6 mirror the IPv4 access rules.

```
ipv6 unicast-routing
interface GigabitEthernet0/0.20
 ip address 192.168.54.65 255.255.255.224
 ipv6 address FD54:ACDE:2026:20::1/64
 ipv6 nd other-config-flag
 ipv6 dhcp server LAN-V6
 ipv6 dhcp pool LAN-V6
 dns-server FD54:ACDE:2026:60::2
 domain-name tlou.local
 ipv6 route ::/0 FD54:ACDE:2026:FFFF::2
```

Why this is appropriate. No ISP-delegated prefix is available in the lab, so a ULA prefix is used. A /64 per VLAN is what SLAAC requires, and stateless DHCPv6 gives DNS without the router holding lease state. Guest is SLAAC-only because it is blocked from the internal DNS server, so a DHCPv6 DNS entry would be unreachable.

Bandwidth constraint. Dual-stack adds no tunnelling or translation overhead on the single WAN link. NAT overload applies only to Guest and Admin, so Arcade and Cafe traffic stays on the LAN.

CR8 shared printer zone. TO-PRINTZONE (IPv4) and TO-PRINTZONE6 (IPv6) on the VLAN 50 sub-interface let only Arcade and Cafe reach the printer, on printing ports and ping. Guest and Admin are denied, matching the Milestone 1 design.

3. Testing Evidence

The results below cover addressing, routing, the CR8 printer zone and Guest isolation over both IPv4 and IPv6.

3.1 Baseline Layer 2 and routing Each access switch places its ports in the correct VLAN and can reach the Services gateway 192.168.54.129. The first ping in each run is lost while ARP resolves, giving 4 of 5.

FIGURE 1: Admin-SW-Vlan

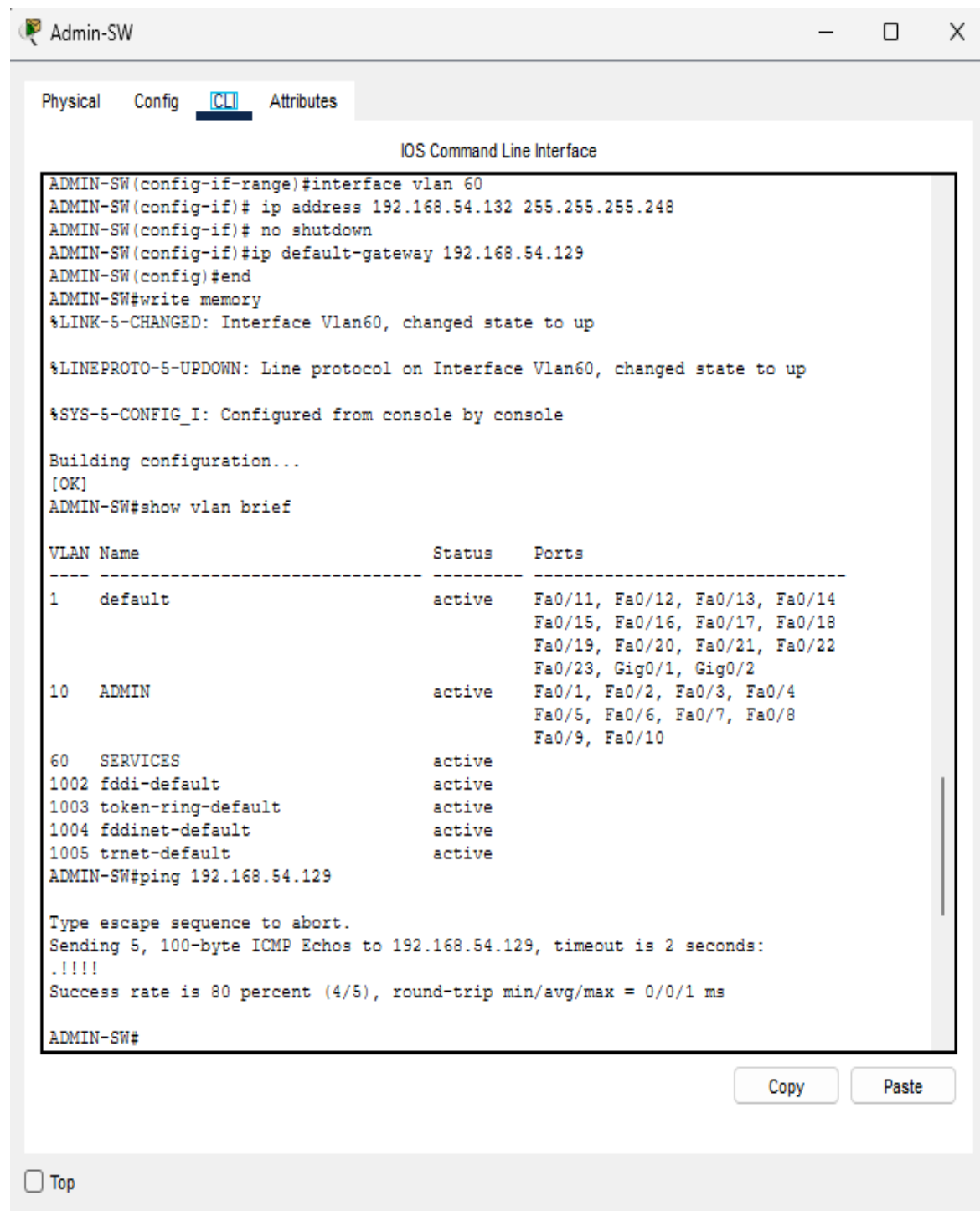
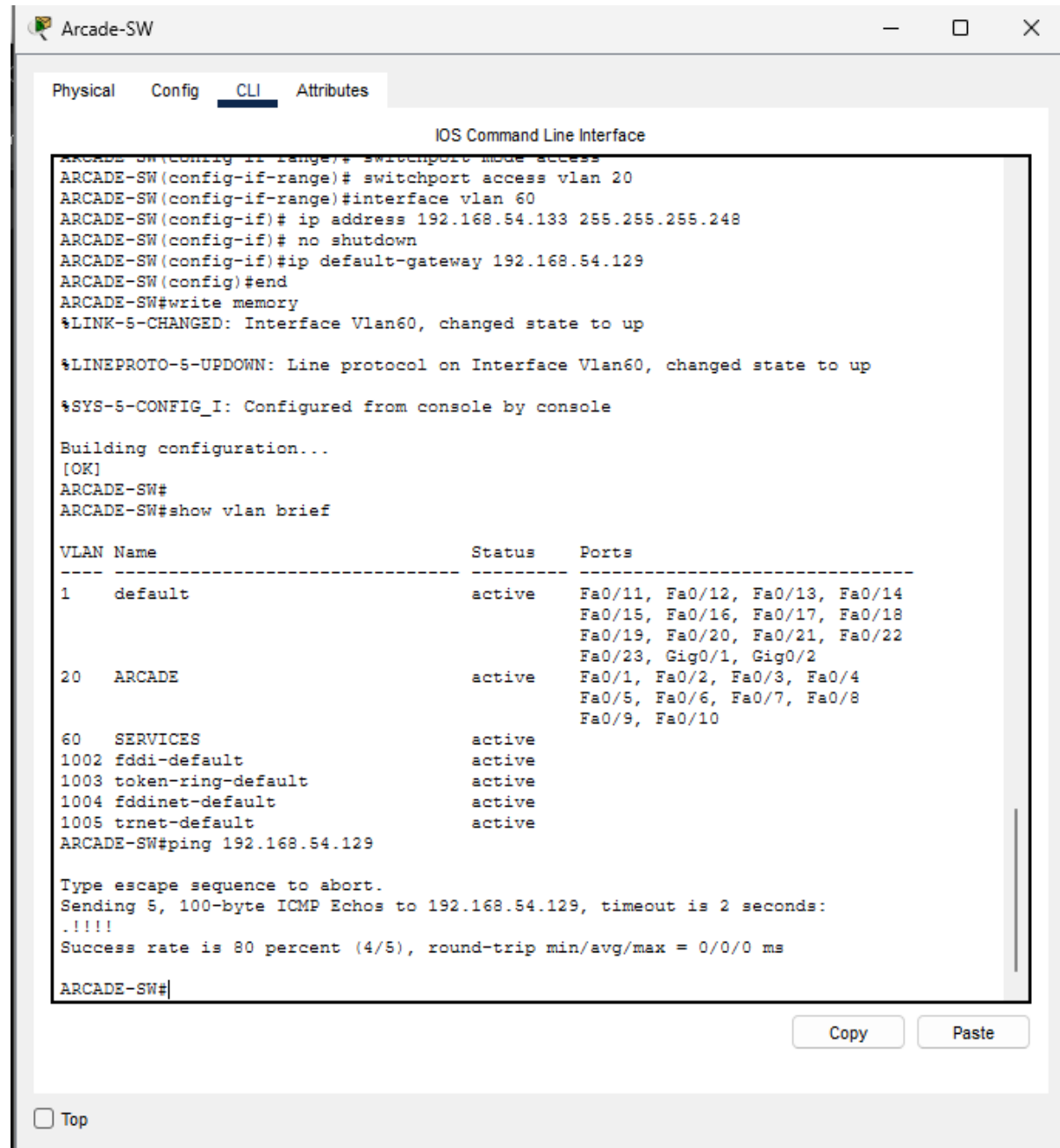


FIGURE 2: Arcade-SW-Vlan



The screenshot shows the CLI interface of a switch named 'Arcade-SW'. The 'CLI' tab is selected. The interface displays the following commands and output:

```
ARCade-SW(config-if-range)# switchport mode access
ARCade-SW(config-if-range)# switchport access vlan 20
ARCade-SW(config-if-range)# interface vlan 60
ARCade-SW(config-if)# ip address 192.168.54.133 255.255.255.248
ARCade-SW(config-if)# no shutdown
ARCade-SW(config-if)# ip default-gateway 192.168.54.129
ARCade-SW(config)# end
ARCade-SW# write memory
%LINK-5-CHANGED: Interface Vlan60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
ARCade-SW#
ARCade-SW# show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Gig0/1, Gig0/2
20 ARCADE	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
60 SERVICES	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

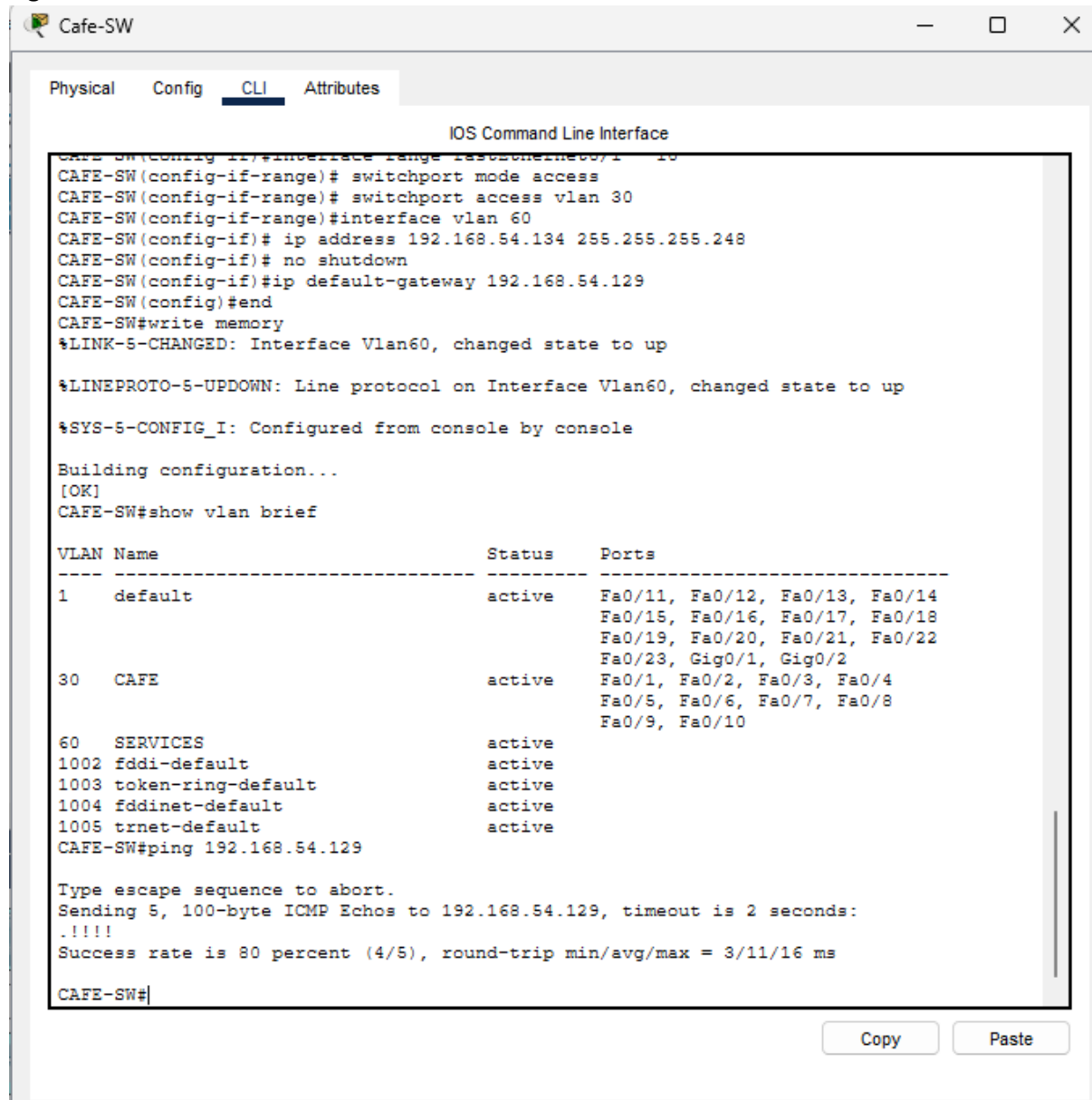
```
ARCade-SW# ping 192.168.54.129

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.54.129, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

ARCade-SW#
```

At the bottom of the window, there are 'Copy' and 'Paste' buttons, and a 'Top' link.

Figure 3: Cafe-SW-Vlan



CORE-SW show vlan brief and show interfaces trunk (VLANs 10 to 60 active, trunks up)

Figure4: Core trunk

The screenshot shows the Core-SW CLI interface with the following content:

Physical Config **CLI** Attributes

IOS Command Line Interface

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/6 (50),
with Switch FastEthernet0/24 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/6 (50),
with Switch FastEthernet0/24 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/6 (50),
with Switch FastEthernet0/24 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/6 (50),
with Switch FastEthernet0/24 (1).

CORE-SW> enable
CORE-SW# show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1
Fa0/2     on        802.1q         trunking    1
Fa0/3     on        802.1q         trunking    1
Fa0/4     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50,60
Fa0/2     1-1005
Fa0/3     1-1005
Fa0/4     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,50,60
Fa0/2     1,10,20,30,40,50,60
Fa0/3     1,10,20,30,40,50,60
Fa0/4     1,10,20,30,40,50,60

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,50,60
Fa0/2     1,10,20,30,40,50,60
Fa0/3     1,10,20,30,40,50,60
Fa0/4     1,10,20,30,40,50,60

CORE-SW#
```

Copy Paste

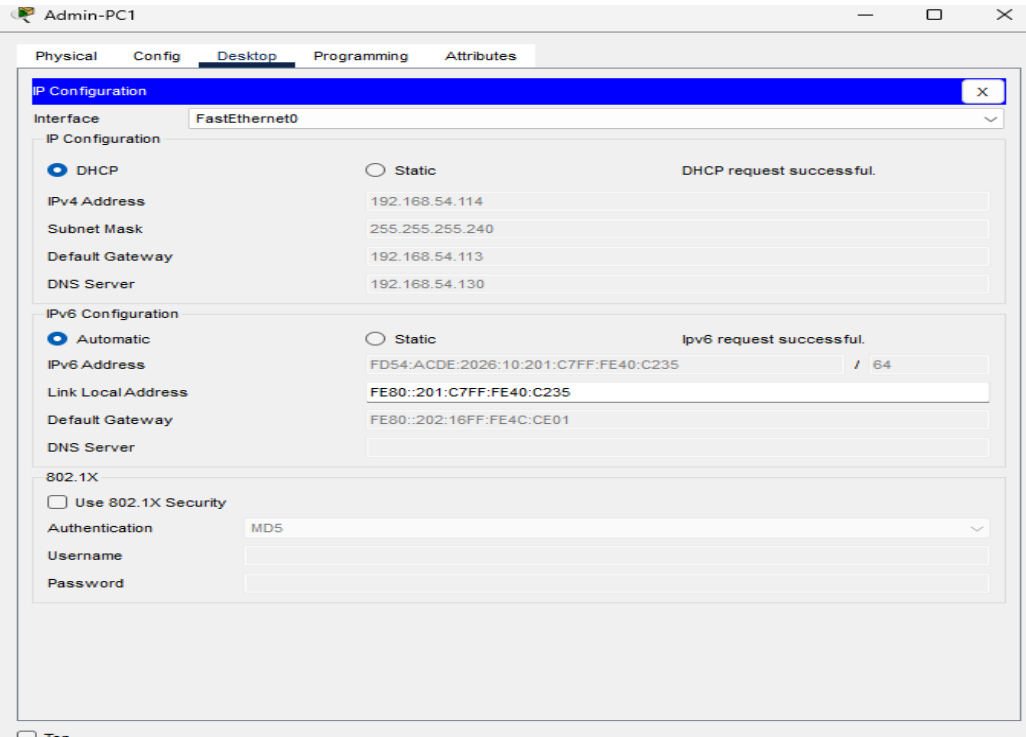
☐ Top

3.2 Dual-stack host addressing and DHCPv6

Hosts receive a DHCPv4 lease and a SLAAC IPv6 address. Admin-PC1 gets 192.168.54.114/28 with gateway 192.168.54.113 and the address FD54:ACDE:2026:10:201:C7FF:FE40:C235. Arcade-PC1 gets

192.168.54.66/27 with gateway 192.168.54.65 and an address in FD54:ACDE:2026:20::/64. The IPv6 default gateway on both is the router link-local address.

Figure 5: DHCP-DNS-Server static addressing



3.3 CR8 printer zone

The router reaches the printer address FD54:ACDE:2026:50::2 with 5 of 5 replies after the printer was given its static IPv6 address. Cafe-PC1 reaches the printer on both stacks with 4 of 4 replies each.

Figure 6: Cafe-PC1 to the printer

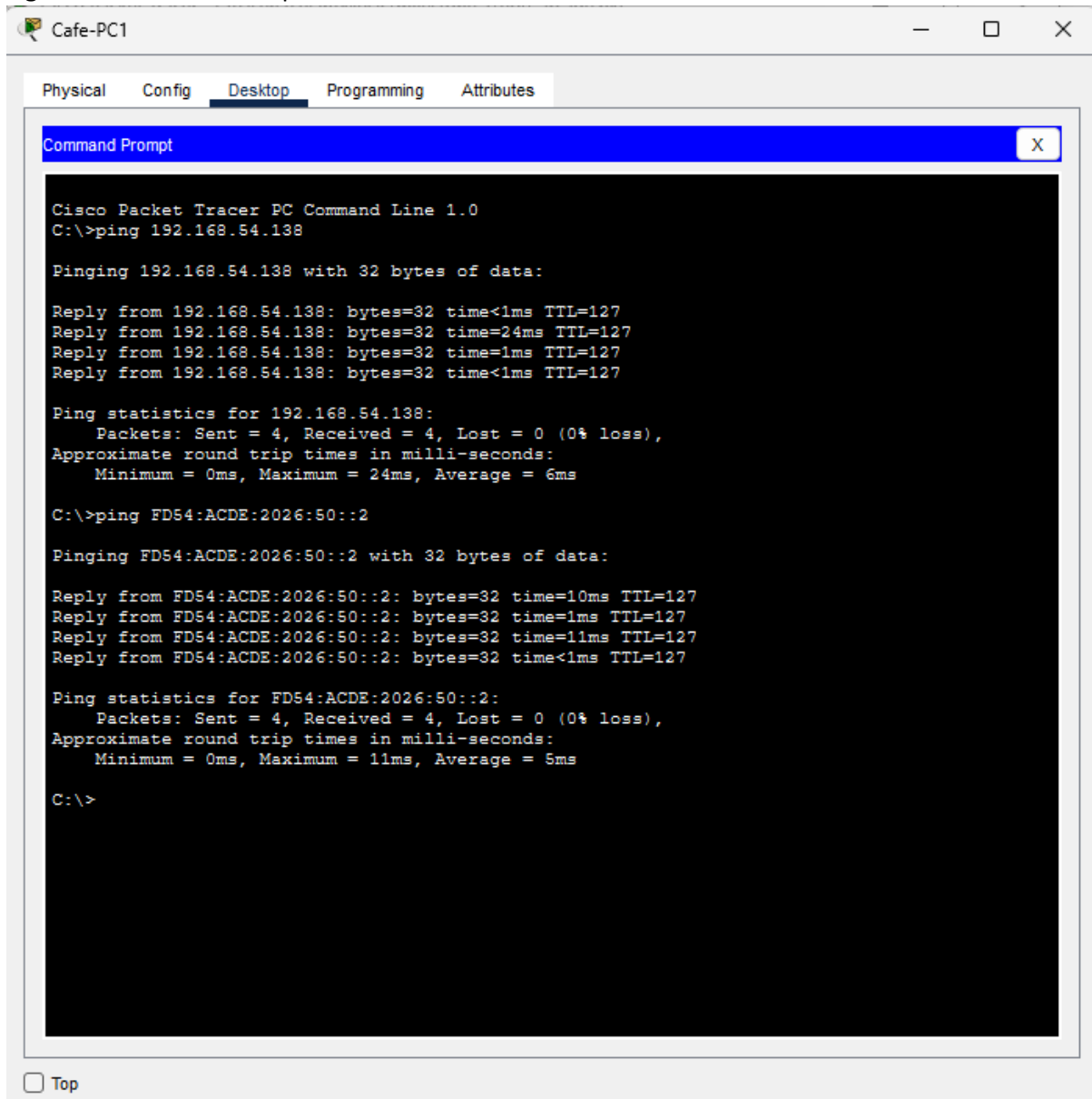


Figure 7: Arcade-to-printer

```
C:\>ping 192.168.54.138

Pinging 192.168.54.138 with 32 bytes of data:

Reply from 192.168.54.138: bytes=32 time<1ms TTL=127
Reply from 192.168.54.138: bytes=32 time<1ms TTL=127
Reply from 192.168.54.138: bytes=32 time<1ms TTL=127
Reply from 192.168.54.138: bytes=32 time=10ms TTL=127

Ping statistics for 192.168.54.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping FD54:ACDE:2026:50::2

Pinging FD54:ACDE:2026:50::2 with 32 bytes of data:

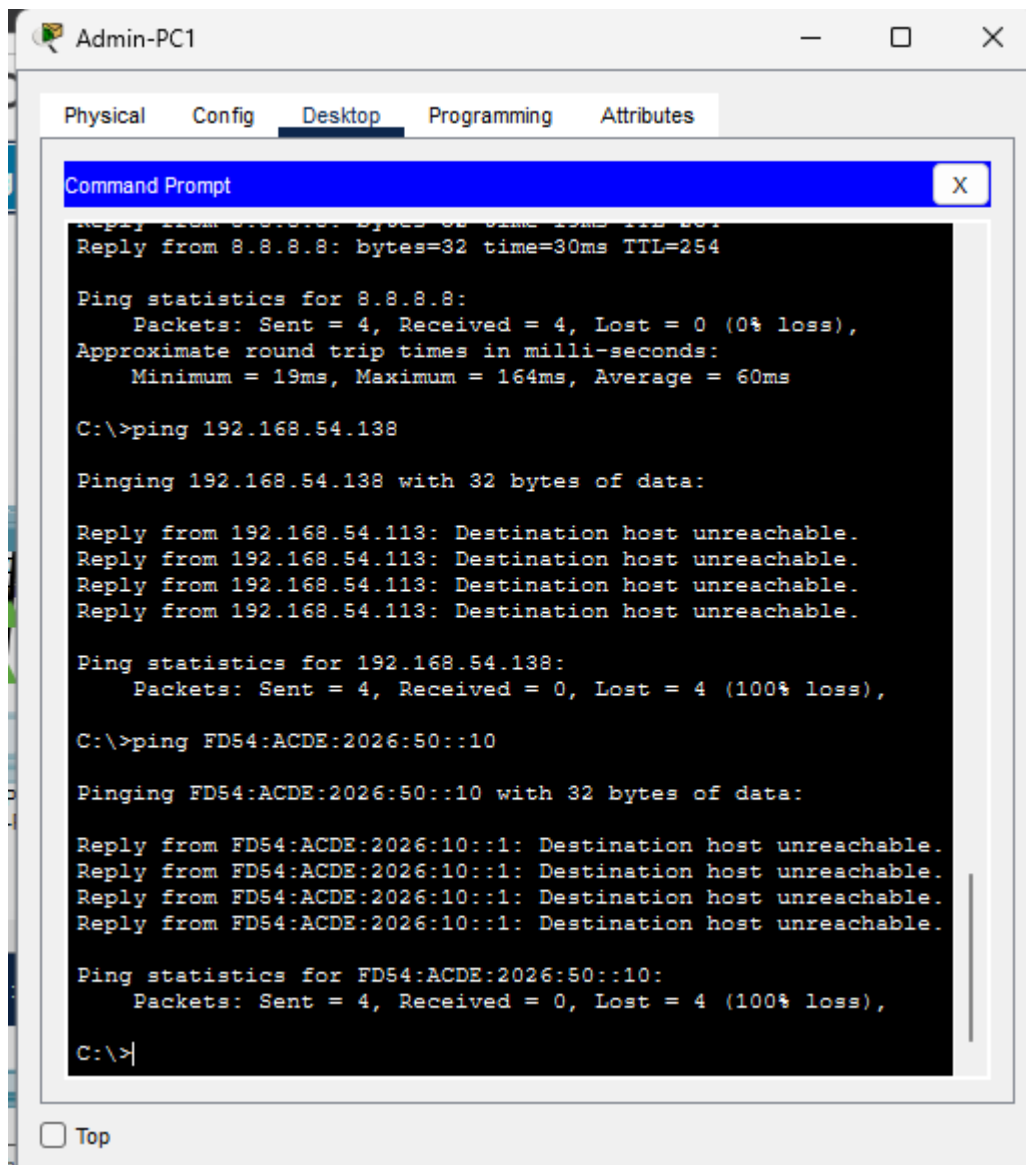
Reply from FD54:ACDE:2026:50::2: bytes=32 time<1ms TTL=127
Reply from FD54:ACDE:2026:50::2: bytes=32 time=27ms TTL=127
Reply from FD54:ACDE:2026:50::2: bytes=32 time<1ms TTL=127
Reply from FD54:ACDE:2026:50::2: bytes=32 time=1ms TTL=127

Ping statistics for FD54:ACDE:2026:50::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 27ms, Average = 7ms

C:\>
```

Admin-PC1 is blocked from the print zone on both stacks. The unreachable replies come from the router Admin interfaces, which confirms the print zone is open only to Arcade and Cafe.

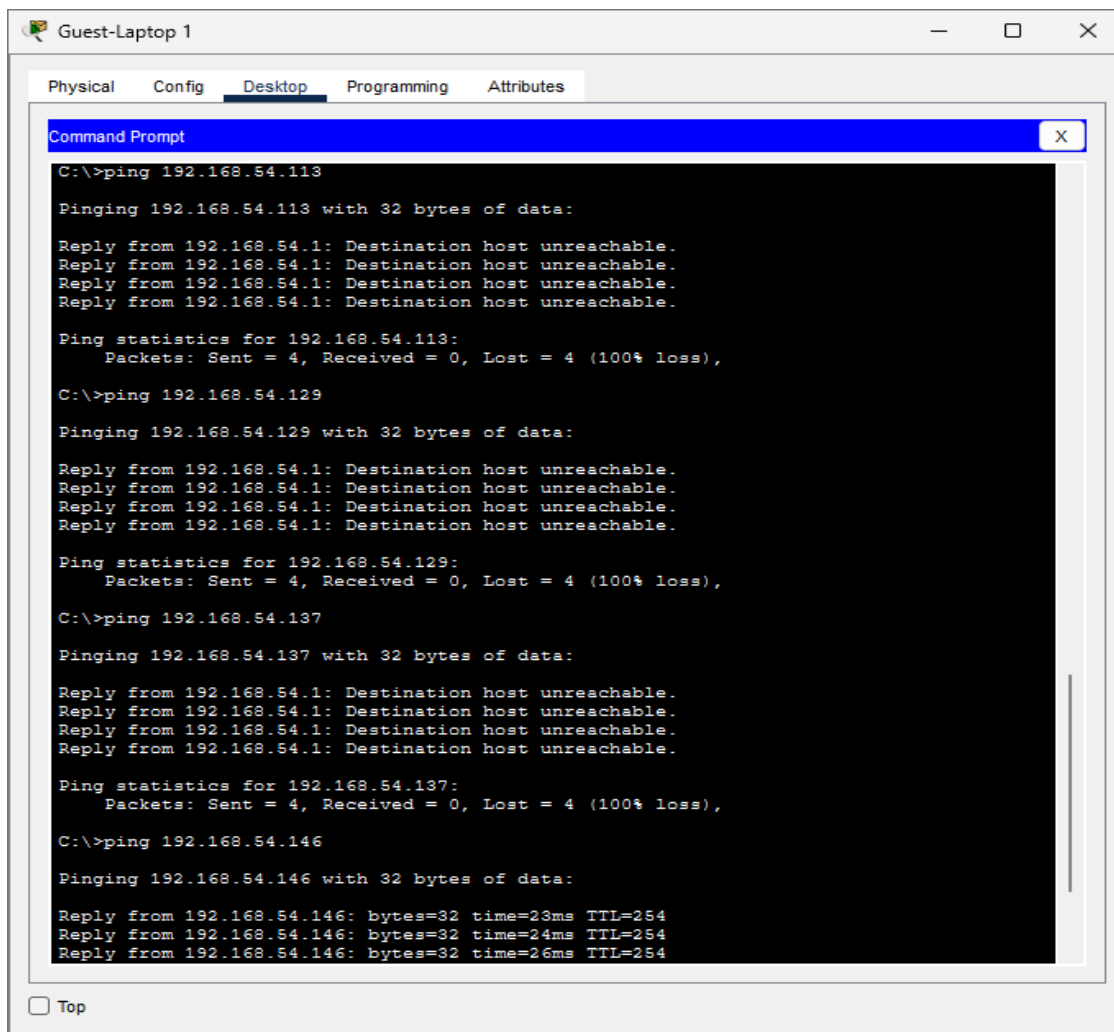
Figure 8: Admin-printer-blocked



3.4 Guest isolation, internet access and NAT

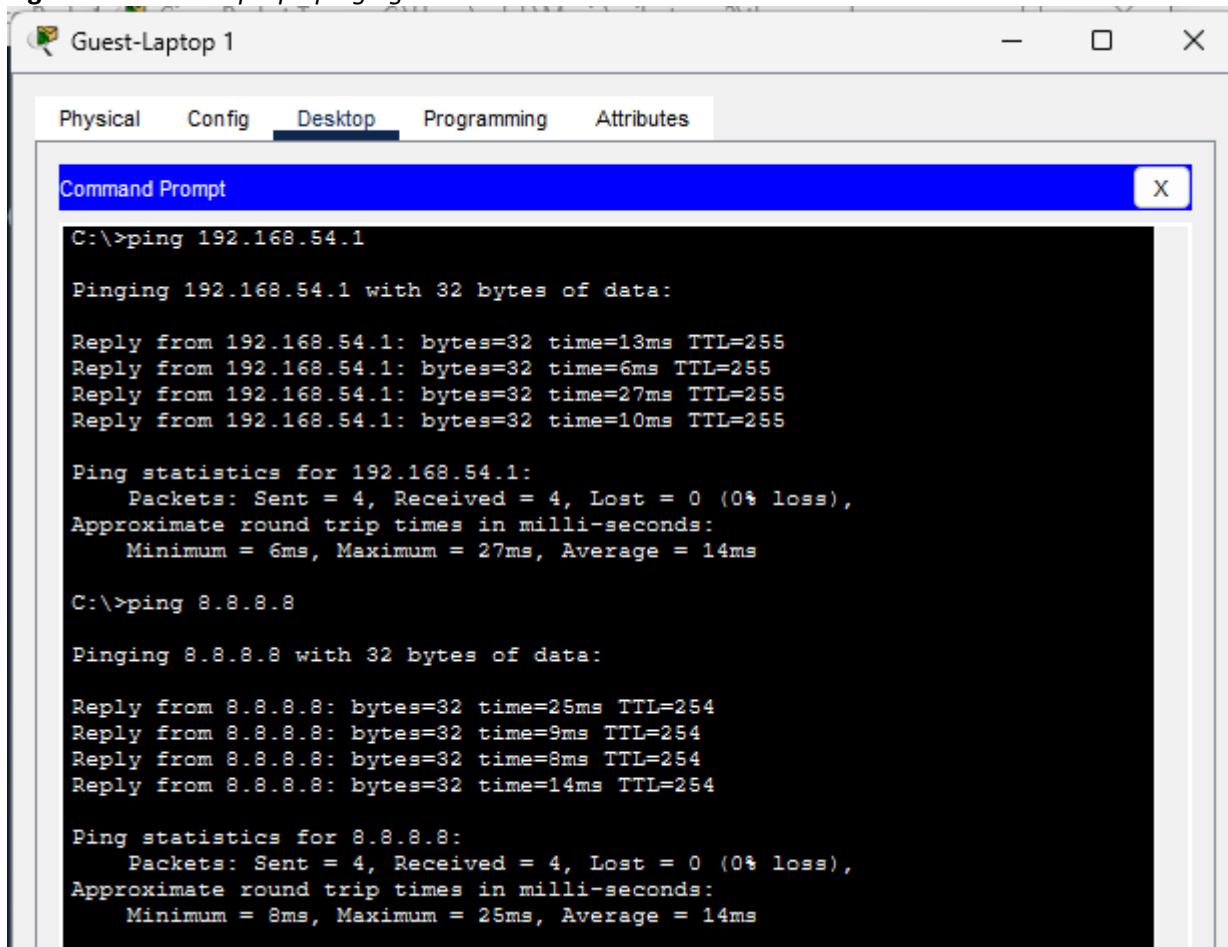
Guest-Laptop1 cannot reach Admin (192.168.54.113), Services (.129) or Print-zone (.137); each ping returns destination unreachable from its gateway 192.168.54.1. It does reach the WAN address 192.168.54.146.

Figure 9: Guest-isolation



Guest reaches its own gateway and the simulated internet host 8.8.8.8, and NAT translates the Guest address 192.168.54.2 to 192.168.54.145.

Figure 10: Guest-Laptop1 ping



The screenshot shows a Windows Command Prompt window titled "Guest-Laptop 1". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" currently selected. The Command Prompt displays the results of two ping commands. The first command is "C:\>ping 192.168.54.1", which shows four successful replies with varying times (13ms, 6ms, 27ms, 10ms) and a TTL of 255. The statistics for 192.168.54.1 show 4 packets sent, 4 received, 0 lost, and an average round trip time of 14ms. The second command is "C:\>ping 8.8.8.8", which also shows four successful replies with times (25ms, 9ms, 8ms, 14ms) and a TTL of 254. The statistics for 8.8.8.8 show 4 packets sent, 4 received, 0 lost, and an average round trip time of 14ms.

```
C:\>ping 192.168.54.1

Pinging 192.168.54.1 with 32 bytes of data:

Reply from 192.168.54.1: bytes=32 time=13ms TTL=255
Reply from 192.168.54.1: bytes=32 time=6ms TTL=255
Reply from 192.168.54.1: bytes=32 time=27ms TTL=255
Reply from 192.168.54.1: bytes=32 time=10ms TTL=255

Ping statistics for 192.168.54.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 27ms, Average = 14ms

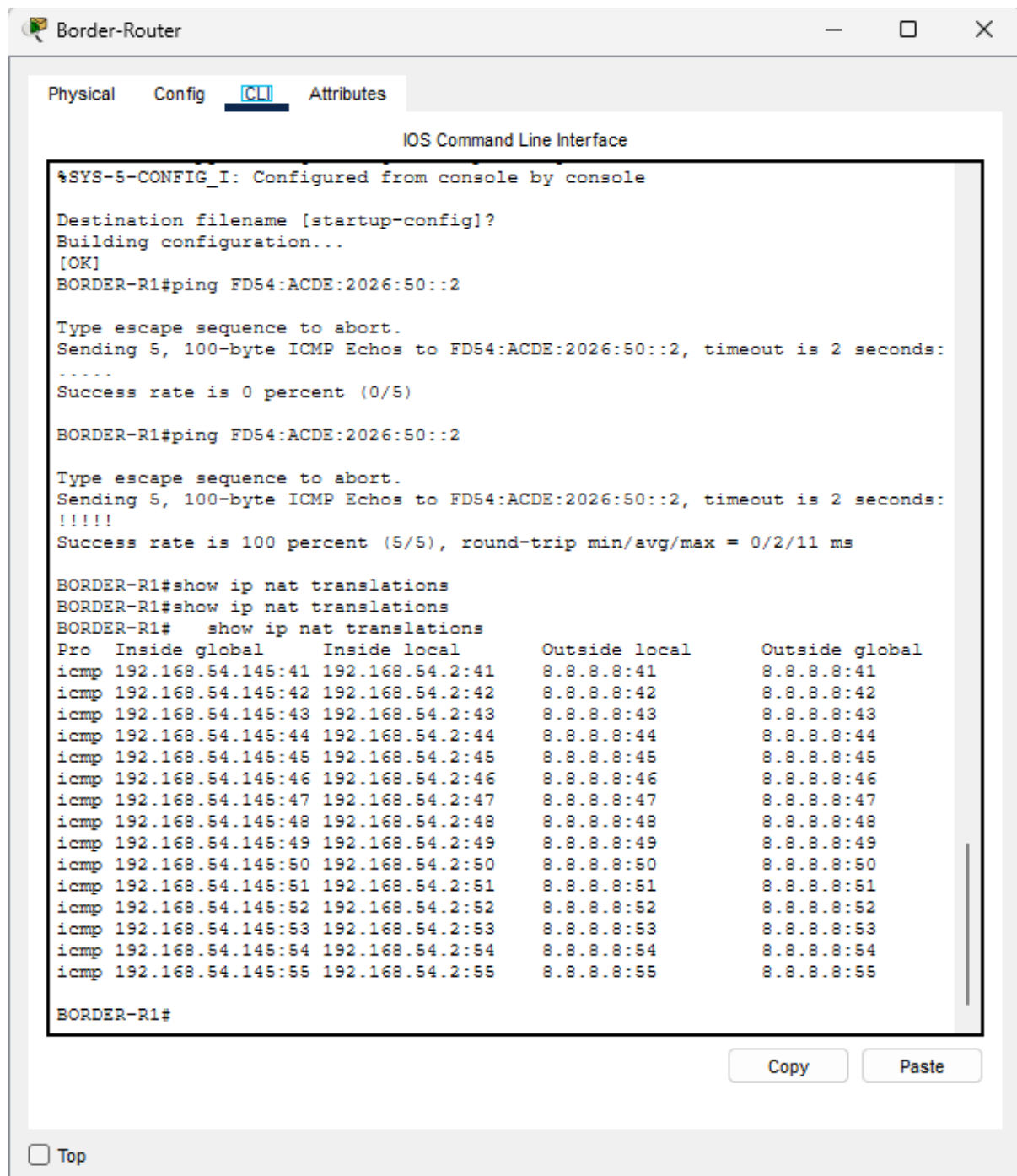
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=25ms TTL=254
Reply from 8.8.8.8: bytes=32 time=9ms TTL=254
Reply from 8.8.8.8: bytes=32 time=8ms TTL=254
Reply from 8.8.8.8: bytes=32 time=14ms TTL=254

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 25ms, Average = 14ms
```

Figure 11: BORDER-R1 show ip nat translations



3.5 Guest isolation over IPv6

Guest-Laptop1 cannot reach FD54:ACDE:2026:10::1 (Admin) or FD54:ACDE:2026:50::2 (Print-zone); both return unreachable from FD54:ACDE:2026:40::1. It reaches its own gateway FD54:ACDE:2026:40::1 with 4 of 4 replies.

Figure 12: Guest-Laptop1 IPv6 tests: Admin and Print-zone blocked

```
C:\> ping FD54:ACDE:2026:10::1

Pinging FD54:ACDE:2026:10::1 with 32 bytes of data:

Reply from FD54:ACDE:2026:40::1: Destination host unreachable.
Reply from FD54:ACDE:2026:40::1: Destination host unreachable.
Reply from FD54:ACDE:2026:40::1: Destination host unreachable.
Reply from FD54:ACDE:2026:40::1: Destination host unreachable.

Ping statistics for FD54:ACDE:2026:10::1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping FD54:ACDE:2026:50::2

Pinging FD54:ACDE:2026:50::2 with 32 bytes of data:

Reply from FD54:ACDE:2026:40::1: Destination host unreachable.
Reply from FD54:ACDE:2026:40::1: Destination host unreachable.
Reply from FD54:ACDE:2026:40::1: Destination host unreachable.
Reply from FD54:ACDE:2026:40::1: Destination host unreachable.

Ping statistics for FD54:ACDE:2026:50::2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping FD54:ACDE:2026:40::1

Pinging FD54:ACDE:2026:40::1 with 32 bytes of data:

Reply from FD54:ACDE:2026:40::1: bytes=32 time=60ms TTL=255
Reply from FD54:ACDE:2026:40::1: bytes=32 time=9ms TTL=255
Reply from FD54:ACDE:2026:40::1: bytes=32 time=9ms TTL=255
Reply from FD54:ACDE:2026:40::1: bytes=32 time=20ms TTL=255

Ping statistics for FD54:ACDE:2026:40::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 60ms, Average = 24ms
```

4. Troubleshooting Summary

Six issues were found and fixed during the build.

1. **Guest could not reach the WAN.** GUEST-IN contained deny ip any 192.168.54.128 0.0.0.127, which covers .128 to .255 including the WAN link. It was replaced with 0.0.0.15 (.128 to .143 only). The first attempt used no 25 and removed nothing because this Packet Tracer does not show sequence numbers, so the old rule was removed by its text.
2. **Stray interface and ACL.** GigabitEthernet0/1.40 and GUEST-OUT were created on the WAN port instead of the trunk on Gi0/0. Both were removed.
3. **Printer unreachable over IPv6.** IPv4 worked, but IPv6 failed, and the router could not reach the printer either, which ruled out the ACL. The printer had no IPv6 address; setting FD54:ACDE:2026:50::2/64 and its gateway fixed it.
4. **Guest could not ping its own IPv6 gateway.** GUEST6-IN denied the whole /48. Permits for the Guest /64 and the WAN /64 were added ahead of the deny.
5. **DHCPv6 is missing.** The design specified stateless DHCPv6 but the router had none. Pool LAN-V6 and the other-config flag were added on VLANs 10, 20 and 30.

6. **Packet Tracer crashed.** The newest. pkt was recovered and the router configuration was intact, but the last fixes had to be reapplied. Versioned saves and configs kept as text in configs/ now protect against a repeat.

Device Configuration Summary

Device	Role	Config file
BORDER-R1	Border router: sub-interfaces, dual-stack, NAT, ACLs, DHCPv6	BORDER-R1.txt
ISP	Simulated ISP with IPv4 and IPv6 loopbacks and a return route to the LAN prefix	ISP.txt
CORE-SW	Core switch: VLANs, trunk to BORDER-R1 and access switches, Guest AP and server ports	CORE-SW.txt
ADMIN-SW	Access switch, VLAN 10	ADMIN-SW.txt
ARCADE-SW	Access switch, VLAN 20	ARCADE-SW.txt
CAFE-SW	Access switch, VLAN 30	CAFE-SW.txt
PRINTZONE-SW	Print-zone switch for the shared printer, VLAN 50	PRINTZONESW.txt
DHCP-DNS-Server	DHCP and DNS services, VLAN 60	server-notes.md